Seo-Young Lee

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RESEARCH INTEREST

My research focuses on human-aligned AI through multimodal representation learning. I develop frameworks that bridge human latent states—such as natural language instructions and tactile signals—with agent behavior. My work spans procedural content generation (PCG), 3D human pose estimation, and zero-shot coordination in both gaming and physical domains.

Keywords: reinforcement learning, multimodal representation learning, PCG, human-AI interaction, human pose estimation

EDUCATION

M.S. in Artificial Intelligence Aug 2023 – Aug 2025

Gwangju Institute of Science and Technology (GIST) · AI융합학과(구 AI대학원)

GPA: 4.20/4.50

- **Thesis:** *Human-Like Procedural Level Generation via Reinforcement Learning with Contrastive Language-State Embedding*
- **Advisor:** *Kyung-Joong Kim*

B.S. in Computer Science and Engineering Mar 2017 – Feb 2023

Dongguk University

(GPA: 3.96/4.5)

PUBLICATIONS

** denotes equal contribution*

[1] **Human-Aligned Procedural Level Generation Reinforcement Learning via Text-Level-Sketch Shared Representation**
In-Chang Baek*, **Seo-Young Lee***, Sung-Hyun Kim, Geum-Hwan Hwang, Kyung-Joong Kim.
IEEE Transactions on Games, Early Access, 2026. DOI: 10.1109/TG.2026.3713793.

[2] **IPCGRL: Language-Instructed Reinforcement Learning for Procedural Level Generation**
In-Chang Baek, Sung-Hyun Kim, **Seo-Young Lee**, Dong-Hyeon Kim, Kyung-Joong Kim.
IEEE Conference on Games (CoG), 2025.

[3] **Multi-Objective Instruction-Aware Representation Learning in Procedural Content Generation Reinforcement Learning**
Sung-Hyun Kim, Geum-Hwan Hwang, In-Chang Baek, **Seo-Young Lee**, Kyung-Joong Kim.
IEEE Conference on Games (CoG), 2026. Accepted; Best Paper Award Nominee.

[4] **Automatic Curriculum Design for Zero-Shot Human-AI Coordination**
Won-Sang You, Tae-Gwan Ha, **Seo-Young Lee**, Kyung-Joong Kim.
IEEE Access, 2025.

[5] **Smart Insole: Predicting 3D human pose from foot pressure**
Isaac Han, **Seo-Young Lee**, Sang-Yeon Park, Ecehan Akan, Yiyue Luo, Kyung-Joong Kim
2nd NeurIPS Workshop on Touch Processing: From Data to Knowledge, 2024.

[6] **Shared Representation for 3D Pose Estimation, Action Classification, and Progress Prediction from Tactile Signals**
Isaac Han, **Seo-Young Lee**, Sang-Yeon Park, Ecehan Akan, Yiyue Luo, Joseph DelPreto, Kyung-Joong Kim
arXiv preprint, 2026.

PROJECTS

Funded Research

Instruction Agent Research based on Natural Language-Reward Function Embedding Jul 2024 – Jun 2025
Principal Investigator (PI) *National Research Foundation (NRF), Korea*

- Authored the proposal for an NRF-funded individual research project and led method development, MuJoCo validation, and interim/final reporting.
- Developed language–reward representations and instruction-conditioned RL; adapted the research to PCGRL, with outcomes including a master's thesis and the collaborative VIPCGRL paper.

HCI + AI for Human-Centered Physical System Design Oct 2023 – Jan 2025
Graduate Researcher *GIST-MIT Project, GIST*

- Contributed to insole sensor fabrication and multi-camera motion-capture setup, including calibration and 3D keypoint extraction; the Smart Insole setup was reused in SCOTTI. [5,6]
- Proposed and implemented SCOTTI's CNN–Transformer encoder for spatiotemporal tactile features supporting pose, action, and progress prediction. [6]

Human Activity Recognition & Object Detection for Art Creation Dec 2022 – Mar 2023
HAR Model Development & Mobile Integration *Samsung C-Lab / Dongguk University*

- Proposed the HAR model pipeline and trained and evaluated the model; contributed to ONNX conversion and inference integration into an Android demo app developed by a teammate.

Academic Projects

PCGRL Research Series: Language-Instructible and Human-Aligned Level Generation **Jan 2025 – Jan 2026**
Academic research series

- **HL-PCGRL:** Designed and implemented contrastive language–state learning, policy conditioning, and human-similarity rewards; conducted experiments, analysis, and thesis writing.
- **VIPCGRL:** Proposed major model architectures, designed shared-representation losses, implemented CLIP models, datasets/samplers, training code and RL integration, and served as a main writer. [1]
- **IPCGRL:** Advised on adapting the NRF method to PCGRL, explored VAE/KL alternatives, and contributed to manuscript writing. [2]

Adaptive Curriculum Learning for Human-AI Zero-Shot Coordination Sep 2023 – Feb 2024
Graduate Researcher

- Implemented a cross-play evaluation pipeline and analyzed curriculum difficulty distributions for zero-shot human–AI coordination in Overcooked. [4]

EXPERIENCE

Cognition and Intelligence Lab, GIST Sep 2025 – Jan 2026
Researcher (Post-Master's)

Artificial Intelligence Lab, Dongguk University Dec 2020 – Mar 2023
Undergraduate Research Assistant

SKILLS

Programming Language: Python, C++
Frameworks: PyTorch, JAX, Flax
Language: Korean, English (TOEIC 825)

Awards & Honors

Master's Research Stipend, National Research Foundation (NRF) (₩12,000,000) Jul 2024 – Jun 2025
Academic Excellence Scholarship, Dongguk University Sep 2022